Department of Computer Science and Engineering and Information Technology

Batch 33 Group A

Software requirements specification

Project Title: Codeforces Companion App

Platform: Mobile Application (Android)

Framework: Flutter

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1. Introduction

1.1 Purpose

The purpose of this mobile application is to help competitive programmers track their progress and improve their problem-solving skills using data from **Codeforces**.

- It solves the problem of manually tracking solved problems, ratings, and contests.
- It is designed for students and programmers who practice competitive programming.
- It is useful because it provides a centralized platform for analytics, recommendations, and reminders.

1.2 Scope

The application will include the following features:

- Login using Codeforces handle
- Display user rating, rank, and statistics
- Show solved problems and submission history
- Daily problem recommendation system
- Contest reminders and notifications
- Bookmark favorite problems

Extra Feature:
- Random problem generator (800–1000 rating range)

Limitations:
- The app will not support direct problem submission
- No web version in this phase
- Limited to Codeforces platform only

1.3 Objectives

- To design a user-friendly mobile application using Flutter
- To integrate Codeforces API for real-time data
- To provide personalized problem recommendations
- To help users improve consistency in practice
- To implement clean UI and smooth user experience

1.4 Target Users

- Competitive programming beginners (800–1200 rating)
- Intermediate programmers preparing for contests
- Students interested in improving problem-solving skills
- Regular users of **Codeforces**

1.5 Definitions, Acronyms, and Abbreviations

- **API** – Application Programming Interface
- **UI** – User Interface
- **UX** – User Experience
- **CP** – Competitive Programming
- **Handle** – Username on Codeforces
- **Rating** – Skill level score on Codeforces

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2. Overall Description

2.1 Product Perspective

The Codeforces Companion App is a standalone mobile application that interacts with the **Codeforces** API to fetch user data, problems, and contest information.

It does not have its own backend server; instead, it relies on:

- Codeforces public API
- Local storage (for bookmarks and preferences)

2.2 Product Functions

The main functionalities of the app include:

- User login via Codeforces handle
- Fetch and display user profile data (rating, rank, stats)
- Track solved problems

- Recommend daily problems based on user rating
- Generate random problems within a selected rating range
- Show upcoming contests and send reminders
- Allow users to bookmark problems
- Display progress analytics (charts and trends)

2.3 User Classes and Characteristics

● General User

- Beginner or intermediate programmer
- Familiar with competitive programming basics
- Uses the app to practice and track progress

● Advanced User (Optional)

- Regular contest participant
- Uses analytics and recommendations for improvement

2.4 Operating Environment

- Platform: Android (primary), iOS (optional)
- Framework: Flutter
- Backend: Codeforces API (no dedicated backend)
- Database: Local storage (Hive / SQLite)
- IDE: Android Studio / VS Code
- Internet: Required for fetching live data

2.5 Design and Implementation Constraints

- Must use Flutter framework
- Dependent on Codeforces API availability
- Requires internet connection for most features
- Limited offline functionality (only bookmarks stored locally)
- No direct integration for code submission

2.6 Assumptions and Dependencies

- The user must have an active internet connection to fetch data from the Codeforces API
- The Codeforces API must be available and responsive
- The user must have a valid Codeforces handle
- The application assumes that the API data format remains consistent
- Notifications depend on device permissions being enabled

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3. Specific Requirements

3.1 Functional Requirements

2.6 Assumptions and Dependencies

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3. Specific Requirements

3.1 Functional Requirements

ID	Requirement Description	Priority
FR1	The system shall allow users to enter their Codeforces handle	High
FR2	The system shall fetch and display user rating, rank, and statistics	High
FR3	The system shall display a list of solved problems	High
FR4	The system shall recommend daily problems based on user rating	High
FR5	The system shall generate random problems within 800–1000 rating range	Medium
FR6	The system shall display upcoming contests	High

FR7	The system shall send reminders for upcoming contests	Medium
FR8	The system shall allow users to bookmark problems	High
FR9	The system shall store bookmarked problems locally	Medium
FR10	The system shall show progress analytics (charts/statistics)	Medium

3.2 Non-Functional Requirements

ID	Requirement Description	Type
NFR1	The app should load within 3 seconds	Performance
NFR2	The app should have a user-friendly interface	Usability
NFR3	The app should handle API failures gracefully	Reliability
NFR4	The app should support smooth navigation and animations	Usability
NFR5	The app should consume minimal device resources	Efficiency
NFR6	The app should ensure data privacy (no sensitive data stored)	Security

3.3 External Interface Requirements

User Interface

- Clean and intuitive mobile UI
- Dashboard with user stats
- Problem list and recommendation screen
- Bookmark section
- Contest reminder screen

Hardware Interface

- Smartphone (Android/iOS)
- Internet connectivity required

Software Interface

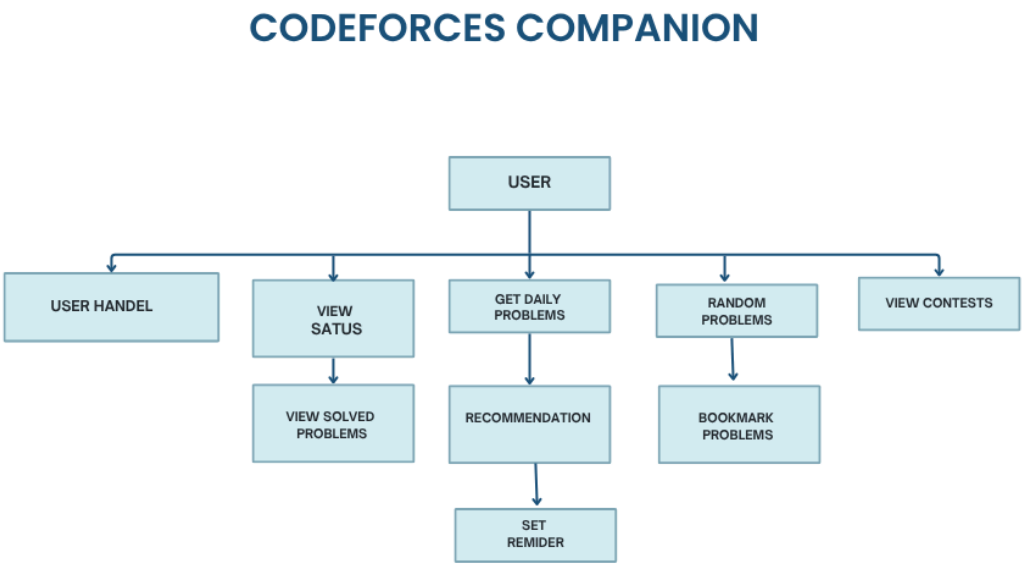
- Codeforces API for fetching user and problem data
- Local database (Hive/SQLite) for storing bookmarks
- Notification system for reminders

4. Project Management

4.1 Development Tools

- Framework: Flutter
- Language: Dart
- API: Codeforces API
- Database: Hive / SQLite
- IDE: Android Studio / VS Code
- Version Control: Git & GitHub

4.2 Timeline

Phase	Task	Duration
Phase 1	Requirement analysis & UI design	2 days
Phase 2	Setup Flutter project & basic UI	2 days
Phase 3	API integration (user data & problems)	3 days
Phase 4	Implement features (recommendation, etc.)	3 days
Phase 5	Testing and debugging	2 days
Phase 6	Final polishing & documentation	2 days

5. System Models